- If we want to combine some n/w just check till what point things are matching
- If after aggregation we cover extra IPs then mention that entry separately and our largest prefix match will take care i.e entry only when they are allocated to someone.
- Misconceptions
(i) addresses should be contiguous
(ii) No of subnets to be combined should be in 2^k

- supernet bits are borrowed from n/w id
- $\boxed{\text{No. of subnets = No of interfaces}}$
- when two routers connect, they use a shared subnet for their connection

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Points to remember:- (Supernetting)
(i) The first address of the block should be divisible by no of addresses of the block.
Eg:- 20.30.40.160
Block contain 32 IP addresses, which is first address of block.
Ans
32) 160 (5
160
×
✓ 20.30.40.160 can be first IP of block

PRIVATE IP:- (Range)
(i) 10.0.0.0 → 10.255.255.255
(ii) 172.16.0.0 → 172.31.255.255
(iii) 192.168.0.0 → 192.168.255.255

- PORT Address:-
To identify process running on the system (16 bit address)
Range: 0 to $2^{16}-1$
⇒ 0 to 65535 port address
(i) 0 to 1023 – predefined ports
(ii) 1024 to 49151 – Registered ports
(iii) 49152 to 65535 – Dynamic port, Epemeral port
http - 80, SMTP - 25, FTP - 21, 20 } predefined port

- Network Mask complement wildcard mask
Eg: 255.255.255.0 → 0.0.0.255
- Host on the n/w:-
AND operation b/w IP and wild card mask
Shortcut: If class C, keep n/w bit 0 and copy host bits as it is
IP: 202.44.89.99 ⇒ Host = 0.0.0.99
- If we need to explicitly configure zero subnet and DBA subnet, then Net ID not possible.
Eg:- IP: 201.44.89.99
Subnet mask: 255.255.255.128
↳ 11111111.11111111.11111111.1 0000000
subnet bits: 0 (zero subnet), 1 (DBA subnet); host bits
- loop back address (127.x.y.z) → Destn IP always
is used to test self computer
- If computer does not have IP address then also it can test itself through DHCP client
(0.0.0.0) → always source IP.
- How to know other's MAC address
↳ ARP protocol (SMAC, DMAC, SIP, DIP, D)
Broadcast signal ⇒ MAC add in return
Eg: OLA, UBER
(i) Unicast MAC Address:-
20:12:13:1A:14:15
0010 0000 → Given to single device
8th bit 0
(ii) Multicast MAC Address
41:12:1A:12:13:15
0100 0001 → given to group/cluster
8th bit 1
(iii) Broadcast MAC Address (DMAC of ARP)
FF:FF:FF:FF:FF:FF
1111 1111
- Transmission Time (for placing data on the channel)
$\boxed{T_T = \frac{F}{B}}$ (Frame size) / Data size; (Bandwidth)
Remember: Kilo = 10^3 ; milli = 10^{-3}
Mega = 10^6 ; micro = 10^{-6}
Giga = 10^9 ; nano = 10^{-9}

- Propagation time (travel in medium)

$$\boxed{P_T = \frac{D}{S}}$$ (Distance)/length of cable
(speed)/velocity of medium

- Round trip time

$$\boxed{RTT = 2 * P_T}$$

Ignore T_T if P_T is large, we take dominate values only

- Total time

$\Rightarrow T_{T(data)} + P_{T(data)} + T_{T(ack)} + P_{T(ack)}$

$\because T_{T(ack)} \downarrow = \frac{\text{Ack size}}{\text{Bandwidth}} \downarrow$

$\therefore T_{T(ack)} \approx$ negligible

& $\boxed{P_{T(ack)} = P_{T(data)}}$

$\therefore \boxed{TT = T_{T(data)} + 2 * P_T}$

- Link Utilization :-

How much time sender is utilizing cable for placing data only.

$$\boxed{\% LU = \frac{T_T}{T_T + 2 * P_T} \times 100\%}$$

- When LU % = 50%

$$\boxed{T_T = 2 * P_T}$$

- Throughput (data rate)

Rate at which user transmit data/frame

$$\boxed{\text{Throughput} = \frac{\text{Data size}}{T_T + 2 * P_T}}$$

$$\boxed{\text{throughput} \leq \text{Bandwidth}}$$

- Another formula for LU %

$$\boxed{\% LU = \frac{\text{throughput}}{\text{Bandwidth}} \times 100\%}$$

- 1 Bit delay means T_T to put 1 bit on cable.